Vincent T. Cook

ROBOTICS & MECHATRONICS

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Aerospace engineering senior with robotics, circuitry design, battery testing, and EV powertrain experience. Apply CAD, structural analysis, MATLAB/Simulink, and dynamics modeling to electromechanical design.

EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

TECHNICAL SKILLS

Design & analysis: SolidWorks, Autodesk Inventor, CAD, stress analysis, simulation and modeling, CFD, thermal testing
Programming & dynamics: MATLAB/Simulink, MotionGenesis, Python, C++, Java | Electromechanical integration

ENGINEERING EXPERIENCE & PROJECTS

SJSU Robotics
Mechanical Designer | Aug 2024–Present
• Designed a transmitter mast with camera and sensor mounts; used stress analysis to reinforce high-load regions and select lightweight materials for lower-load sections.

FIRST Robotics Team 3925
Electrical Lead — High School Robotics | 2020–2022
• Designed circuitry and tested batteries for the team's robot.

SAE Spartan Racing
EV Powertrain Intern | Aug 2023–May 2025
• Created 3D CAD and assembly instructions for an electric motor test bench; integrated mechanical/electrical components and coordinated engineering/manufacturing requirements with teammates.

Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Senior Project
Aerospace Engineering Team Member — In Progress | 2026–Present
• Contribute wing-model designs and yearlong planning for a hybrid battery-electric/hydrogen fuel-cell model aircraft.
• Planned: CAD, CFD/thermodynamics, construction/testing, battery/fuel-system electronics, and commercial-scale modeling; team flight target spring 2027.

BROADSWORD | Reusable Rocket Concept
CAD & Programming Subteam Lead | Sep–Dec 2023
• Developed landing-leg CAD and trajectory-angle-versus-time simulation plots for a reusable rocket concept; coordinated CAD/programming subteams, delegated tasks, and provided design feedback.

NASA L'SPACE Mission Concept Academy
Mechanical Engineering Team Member — Educational Program | Fall 2026–Present
• Help plan science approaches and evaluate mechanical components for a robotic lunar surface payload mission concept.

SELECTED COURSEWORK

• AE 138 / AE 30 | Modeled 3D rigid-body Frisbee flight in MATLAB/MotionGenesis; coded missile launch trajectories.
• AE 157 | Applied MATLAB and Simulink to control-system coursework.
• AE 20 | Created high-level motor designs in introductory CAD coursework.

WORK EXPERIENCE

PayPal Stadium | Guest Representative | 2026 Soccer Season
• Worked San Jose Earthquakes and Bay FC women's team matches.

The Tech Interactive | Educator I | 2024–2025
• Taught STEM labs/design challenges and explained technical concepts to diverse learners.

Self-Employed | Private Swim Instructor | 2023
• Scheduled clients and tailored swim lessons to individual skills and goals.

YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021–2024
• Coached swimmers, led practices, and adapted instruction for children with disabilities.